**Supporting Information for**

# Sulfate Radical-Mediated Degradation of Sulfadiazine by $CuFeO_2$ Rhombohedral Crystal-Catalyzed Peroxymonosulfate: Synergistic Effects and Mechanisms

*Submitted by*

*Yong Feng* [†], *Deli Wu* [‡], *Yu Deng* [†], *Tong Zhang* [†], *and Kaimin Shih* [†,]*

[†]Department of Civil Engineering, The University of Hong Kong, Pokfulam Road, Hong Kong, People's Republic of China

[‡]State Key Laboratory of Pollution Control and Resources Reuse, School of Environmental Science & Engineering, Tongji University, Shanghai 200092, People's Republic of China.

* To whom correspondence should be addressed:

Dr. Kaimin Shih

Phone: +852-2859-1973

Fax: +852-2559-5337

E-mail: kshih@hku.hk (K. Shih)

Fourteen pages: Five text, four tables, and fourteen figures.

## Section S1. Catalyst preparation

***Text S1. Synthesis of $CuFe_2O_4$ particles:*** In a typical procedure, 0.02 M $Fe(NO_3)_3 \cdot 9H_2O$ and 0.01 M $Cu(NO_3)_2 \cdot 9H_2O$ were dissolved in 200 mL of purified water and the solution was stirred and heated at 60 ℃ for 2 h. Then, 0.03 M citric acid was added into the solution, and the obtained homogeneous system was stirred again for 2 h at the same temperature. Finally, the temperature was increased to 85 ℃ to evaporate water, and the resulting complex gel was heated at 400 ℃ for 2 h.

## Section S2. Analytical procedures

***Text S2. SDZ analysis:*** Separation was carried out on a Waters BEH™ C18 column (50 mm × 2.1 mm, 1.7-$\mu$m particle size) at 50 ℃ at a flow rate of 0.4 mL $min^{-1}$ with an injection volume of 10 $\mu$L. The mobile phase was a mixture of methanol and water (with 0.1% (v/v) formic acid as the additive) at a volume ratio of 5%:95%. The calibration curve for SDZ ranged from 0.4 to 10 $\mu$M and exhibited a good linear correlation ($R^2$ = 0.999). The detection limit (S/N = 3) for SDZ was 0.11 $\mu$M.

***Text S3. Degradation product analysis:*** Chromatographic separation of the degradation products was conducted with binary gradient mobile phases of A (water with 0.01% (v/v) formic acid as additive) and B (methanol) at a volume ratio of 95%:5%. Separation was performed on the same column at 50 ℃ at a flow rate of 0.4 mL $min^{-1}$ with an injection volume of 10 $\mu$L. The MS was operated in full-scan, selected ion recording, and multiple reaction monitoring modes. Possible degradation products were first identified by operating the MS in the full-scan mode. Multiple reaction monitoring methods for these products were then established with their corresponding standard reagents. Parameters such as collision energy and cone voltage were optimized for each compound, with the most abundant daughter ion recorded in each case. Other parameters for MS included a source temperature of 120 ℃, a desolvation temperature of 350 ℃, a desolvation gas flow of 400 L $h^{-1}$, a cone gas flow of 50 L $h^{-1}$, and capillary voltage of 0.50 kV. Argon (99.999%) was used as the collision gas, and its pressure in the collision cell was kept at $3.0 \times 10^{-3}$ mbar; nitrogen (≥ 99.995%) was used as the cone and desolvation gas. The full-scan MS data were recorded by scanning from an m/z of 40 to an m/z of 300.

**Section S3. Characterization of catalysts**

***Text S4. Crystallinity and morphology:*** In the X-ray powder diffraction patterns of the $CuFeO_2$ RCs (Figure S1a) and $CuFe_2O_4$ (Figure S1b), all of the diffraction peaks could be exclusively indexed to the known rhombohedral $CuFeO_2$ (PDF # 075-2146) and $CuFe_2O_4$ (PDF # 077-0010), respectively, indicating the prepared catalysts had high purity. The pattern of $Cu_2O$ (Figure S1c) and $Fe_2O_3$ (Figure S1d) matched well with the known cubic $Cu_2O$ (PDS # 005-0667) and α-$Fe_2O_3$ (PDF # 79-1741) and no other peaks could be observed, suggesting that $Cu_2O$ and $Fe_2O_3$ had high purity. The scanning electron microscope image of the $CuFeO_2$ RCs shows that rhombohedral crystals were synthesized and appeared as the main form (Figure S2a). Figure S2b displays the selected area electron diffraction pattern of the $CuFeO_2$ RCs along the [010] zone axis. Based on the high-resolution transmission electron microscopy image (Figure S2c), the interplanar distance was ~0.577 nm, which corresponds to the (003) plane of the rhombohedral $CuFeO_2$ RCs.

***Text S5. Procedures for ATR-FTIR analysis of catalysts during catalytic PMS decomposition:*** The spectra were obtained with a Nicolet Fourier transform infrared spectrometer (Magna-IR 750) equipped with a Universal ATR accessory. Purified water was used to identify background noise. The spectra of the catalysts were calibrated by subtracting the spectrum of purified water during the scanning processes. Details are listed as follows: First, 50 mg of catalyst was mixed with 10 mL purified water (pH 6.8) or PMS solution (20 mM; pH 6.8). After a reaction time of 5 min, the solid particles from the suspensions were scanned in the wavenumber range of 800 to 4000 cm$^{-1}$ at a resolution of 4 cm$^{-1}$.

Table S1 Properties of catalysts used in this study

| catalysts | specific surface area (m$^2$ g$^{-1}$) | average particle size ($\mu$m) | $pH_{pzc}$ |
|---|---|---|---|
| CuO | 1.49 | 0.32 | 9.5 |
| $Cu_2O$ | 4.22 | 0.40 | 5.2 |
| $Fe_2O_3$ | 5.17 | 0.45 | 7.8 |
| $Fe_3O_4$ | 5.40 | 0.30 | 6.5 |
| $CuFeO_2$ RCs | 1.15 | 0.42 | 7.2 |
| $CuFe_2O_4$ | 22.87 | 0.28 | 7.6 |

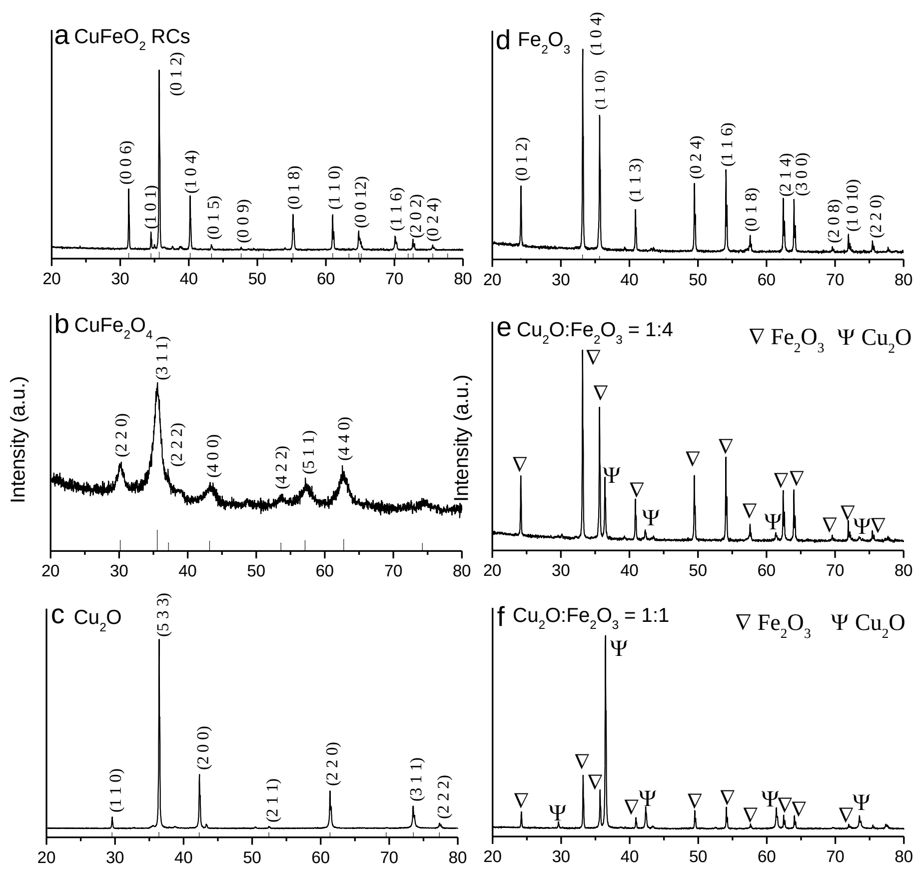



Figure S1. X-ray powder diffraction patterns of $CuFeO_2$ RCs (a), $CuFe_2O_4$ (b), $Cu_2O$ (c), $Fe_2O_3$ (d), $Cu_2O$–$Fe_2O_3$ mixture (1:4) (e), and $Cu_2O$–$Fe_2O_3$ mixture (1:1) (f).

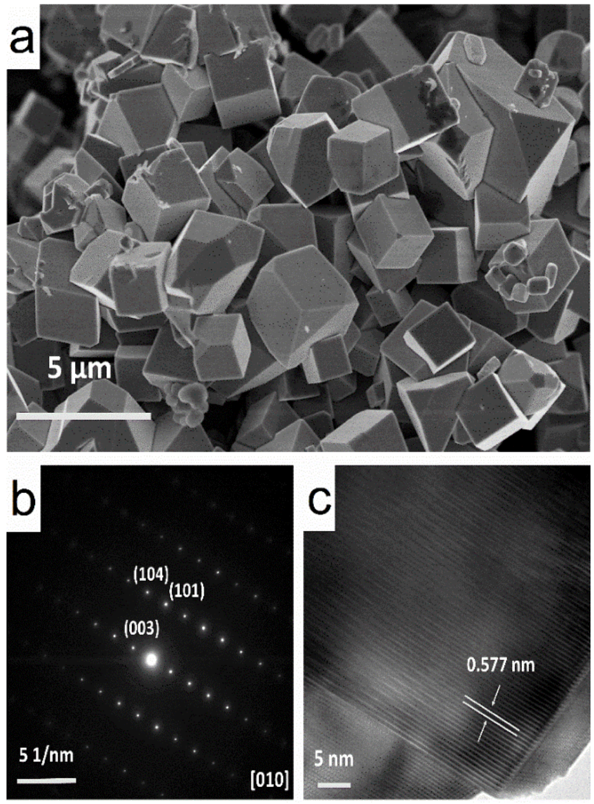



Figure S2. Scanning electron microscope (a), selected area electron diffraction (b), and high-resolution transmission electron microscopy (c) images of $CuFeO_2$ RCs.

**Section S4. Reactivity of catalysts**

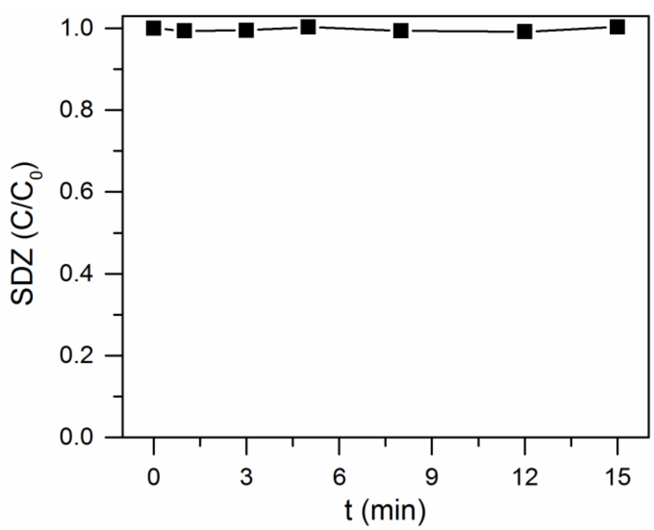



Figure S3. SDZ concentration in $CuFeO_2$ suspension. Conditions: [SDZ] = 8.0 $\mu$M, [$CuFeO_2$ RCs] = 0.1 g $L^{-1}$, and 10 mM phosphate buffer at pH 6.8.

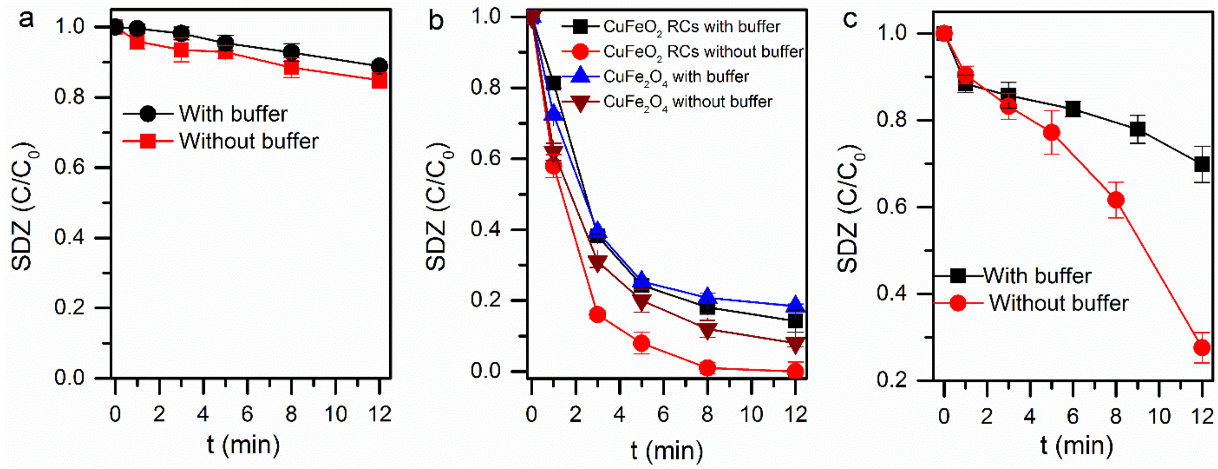



Figure S4. Effect of phosphate buffer (10 mM) on catalytic reactivity of $Cu_2O$ (a), $CuFeO_2$ RCs (b), and CuO (c). Conditions: [SDZ] = 8.0 $\mu$M, [PMS] = 33.0 $\mu$M, [CuO] = [$Cu_2O$] = 0.1 g $L^{-1}$, and pH 6.8.

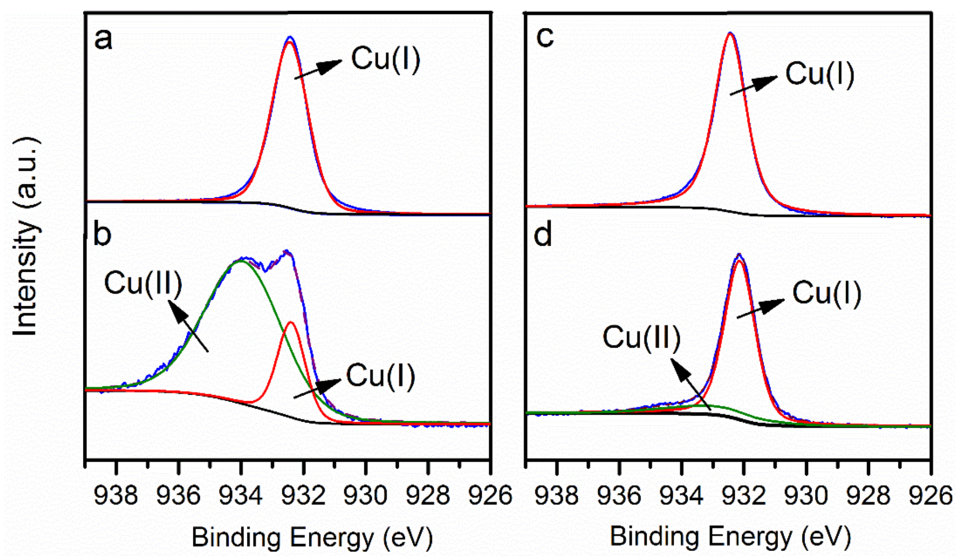



Figure S5. XPS spectra for Cu 2p of the fresh $Cu_2O$ (a), used $Cu_2O$ (b), fresh $CuFeO_2$ RCs (c), and used $CuFeO_2$ RCs (d). Contacting time with PMS were 40 min and 16 min for $Cu_2O$ and $CuFeO_2$ RCs, respectively.

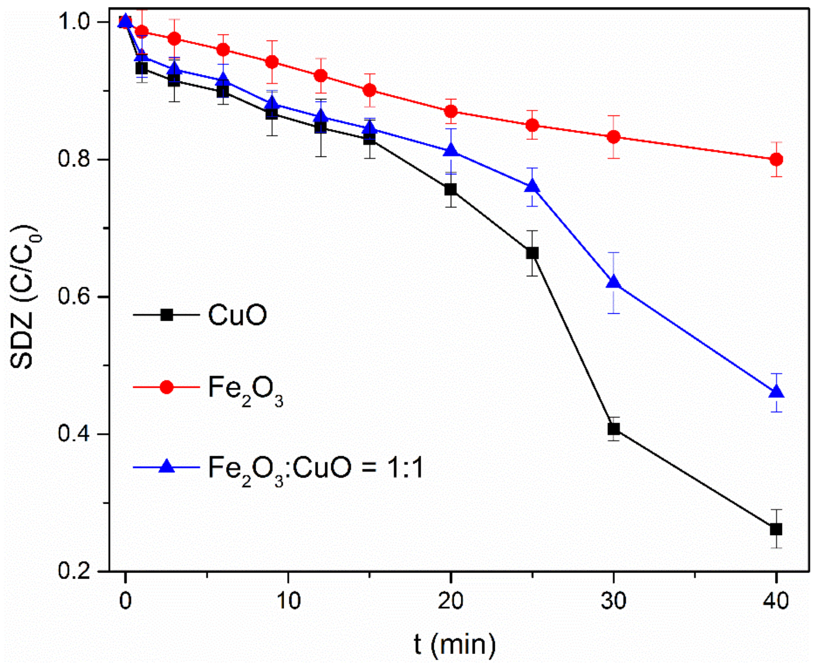



Figure S6. SDZ degradation by PMS with different catalysts. Conditions: [SDZ] = 8.0 $\mu$M, [PMS] = 33.0 $\mu$M, [Catalysts] = 0.05 g $L^{-1}$, and 10 mM phosphate buffer at pH 6.8.

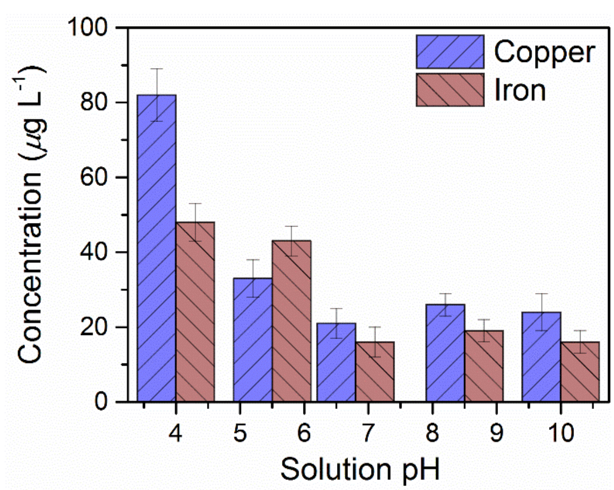



Figure S7. Concentrations of leached copper and iron in $CuFeO_2$ RCs–PMS oxidation under different solution pH values. Conditions: [SDZ] = 8.0 $\mu$M, [PMS] = 33.0 $\mu$M, [$CuFeO_2$ RCs] = 0.1 g $L^{-1}$, and t = 16 min.

## Section S5. Identification of radicals

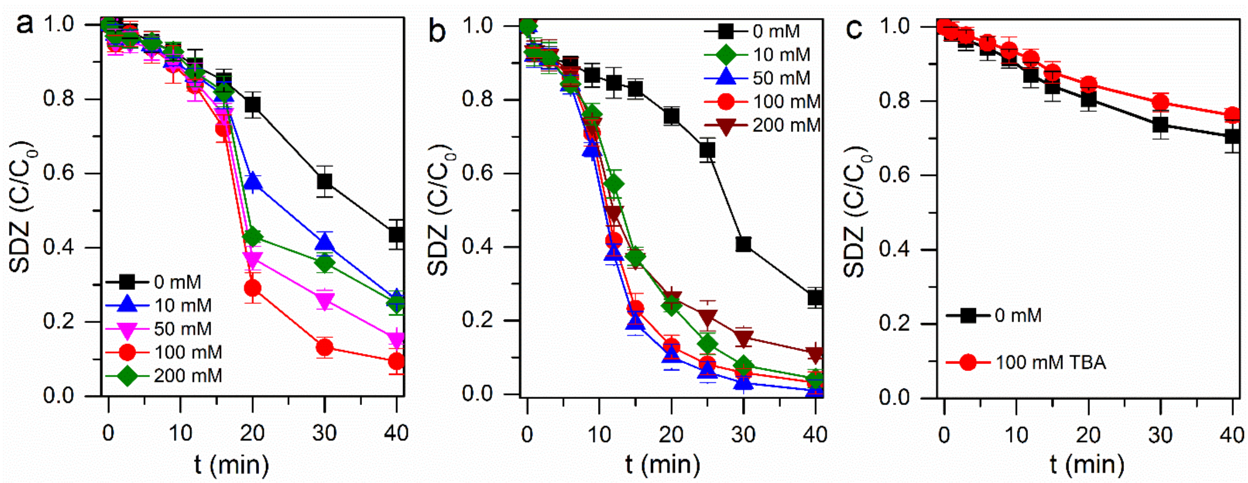



Figure S8. Effect of TBA on SDZ degradation in $Cu_2O$–PMS (a), CuO–PMS (b), and $Fe_2O_3$–PMS (c). Conditions: [TBA] = 100 mM, [SDZ] = 8.0 $\mu$M, [$Cu_2O$] = [CuO] = [$Fe_2O_3$] = 0.05 g $L^{-1}$, [PMS] = 33.0 $\mu$M, and 10 mM phosphate buffer at pH 6.8.

Table S2 Effect of TBA on the pseudo-first-order degradation constants for SDZ degradation in $Cu_2O$–PMS oxidation [a]

| | TBA concentration (mM) | | | | |
|---|---|---|---|---|---|
| | 0 | 10 | 50 | 100 | 200 |
| first 20 min (0–20 min) | 0.012 (0.975) | 0.011 (0.969) | 0.012 (0.915) | 0.013 (0.887) | 0.010 (0.911) |
| final 20 min (20–40 min) | 0.029 (0.999) | 0.045 (0.981) | 0.060 (0.913) | 0.079 (0.885) | 0.042 (0.841) |

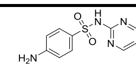
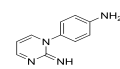
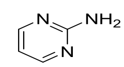
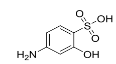
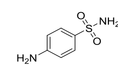
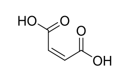
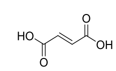
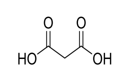
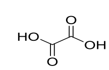
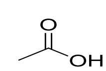
[a] The numbers in the parentheses represent the R-squared values.

Table S3 Rate constants for alcohol oxidation by radicals

| reactions | rate constants ($M^{-1}$ $s^{-1}$) | reference |
|---|---|---|
| $\cdot OH + TBA \rightarrow$ products | $6.0 \times 10^8$ | 1 |
| $\cdot OH +$ ethanol $\rightarrow$ products | $(1.2–2.8) \times 10^9$ | 1 |
| $SO_4^{\cdot-} + TBA \rightarrow$ products | $4.0 \times 10^5$ | 1 |
| $SO_4^{\cdot-} +$ ethanol $\rightarrow$ products | $(1.6–7.7) \times 10^7$ | 2 |

## Section S6. Determination of degradation products

Table S4. Products of SDZ degradation by $CuFeO_2$ RCs–PMS oxidation

| product | parent ion (m/z) | R.T. [c] (min) | fragment ion (m/z) | collision energy (eV) | molecular formula | difference from SDZ | proposed structure |
|---|---|---|---|---|---|---|---|
| SDZ | 251[a] | 1.84 | 156[d], 108, 96, 92[e] | 15 | $C_{10}H_{10}N_4O_2S$ | |  |
| (A) Hydroxyl SDZ | 267[a] | 0.33/1.41 | - | - | $C_{10}H_{10}N_4O_3S$ | +16 | - |
| (B) 4-[2-imino-pyrimidine-l(2H)-yl]aniline | 187[a] | 0.54 | 145[d], 118 92[e], 81 | 25 | $C_{10}H_{10}N_4$ | -64 |  |
| (C) 2-aminopyrimidine | 94[a] | 0.57 | 79[d], 69[e] 53 | 20 | $C_4H_5N_3$ | -155 |  |
| (D) Hydroxyl sulfanilic acid | 191[a] | 0.39 | 177, 123[b], 107[e] | 10 | $C_6H_7NO_4S$ | -60 |  |
| (E) Sulfanilamide | 173[a] | 0.52 | 156[d], 141[e] | 5 | $C_6H_8N_2O_2S$ | -78 |  |
| (F) Maleic acid | 115[b] | 0.50 | 107, 98[e], 71[d], 48 | 10 | $C_4H_4O_4$ | -136 |  |
| (G) Fumaric acid | 115[b] | 0.53 | 98[e], 71[d], 47 | 10 | $C_4H_4O_4$ | -136 |  |
| (H) Malonic acid | 103[b] | 0.54 | 63, 59[d], 45[e] | 10 | $C_3H_4O_4$ | -148 |  |
| (I) Oxalic acid | 89[b] | 0.51 | 59[e], 45, 43[d] | 10 | $C_2H_2O_4$ | -162 |  |
| (J) Acetic acid | 61[a] | 0.36 | 58[e], 44[d] | 10 | $C_2H_4O_2$ | -190 |  |

[a, b] Numbers represent the mass of protonated and deprotonated molecular ions, respectively. [c] R.T. indicates the retention time of each degradation product in the UPLC-MS/MS system. [d] The most abundant fragment ion used in the multiple reaction monitoring mode as the daughter ion. [e] The second most abundant fragment ion.

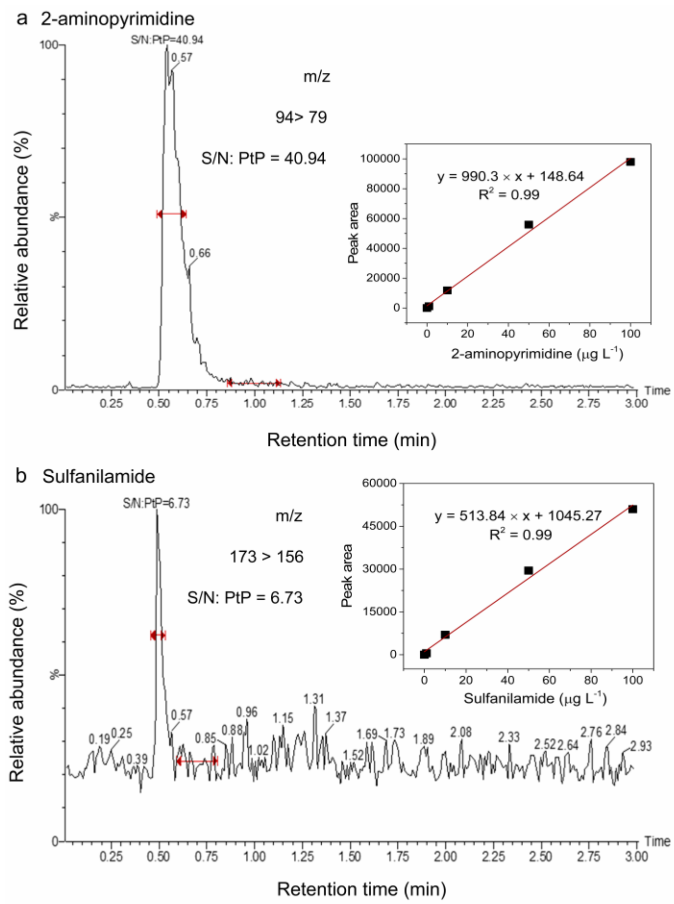



Figure S9. UPLC-MS/MS multiple reaction monitoring chromatograms of 2-aminopyrimidine (a) and sulfanilamide (b) after reacting for 10 min. Insets show the standard curves of these two products. Conditions: [SDZ] = 8.0 $\mu$M, [$CuFeO_2$ RCs] = 0.1 g $L^{-1}$, [PMS] = 33.0 $\mu$M, and 10 mM phosphate buffer at pH 6.8.

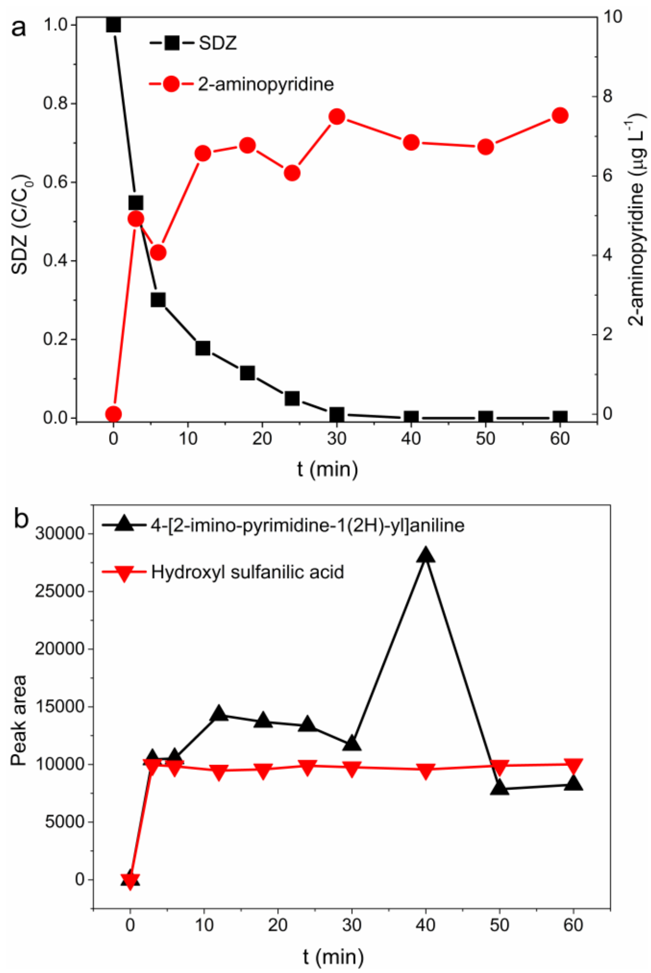



Figure S10. Concentration of SDZ and its degradation products versus reaction time. Conditions: [SDZ] = 8.0 $\mu$M, [$CuFeO_2$ RCs] = 0.1 g $L^{-1}$, [PMS] = 33.0 $\mu$M, and 10 mM phosphate buffer at pH 6.8.

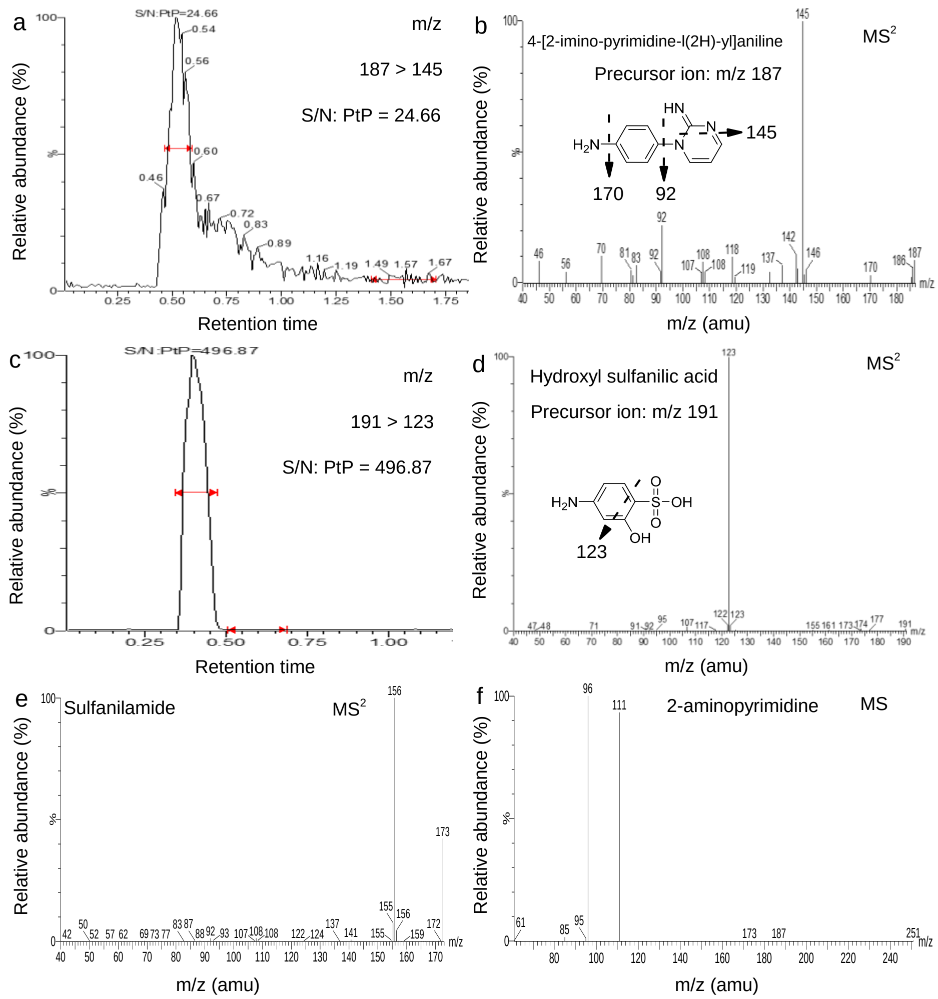



Figure S11. UPLC-MS/MS multiple reaction monitoring chromatograms of 4-[2-imino-pyrimidine-1(2H)-yl]aniline (a) and hydroxyl sulfanilic acid (c). UPLC-MS/MS product ions of 4-[2-imino-pyrimidine-1(2H)-yl]aniline (b), hydroxyl sulfanilic acid (d), sulfanilamide (e), and 2-aminopyrimidine (f).

## Section S7. Reusability and stability of the $CuFeO_2$ RCs

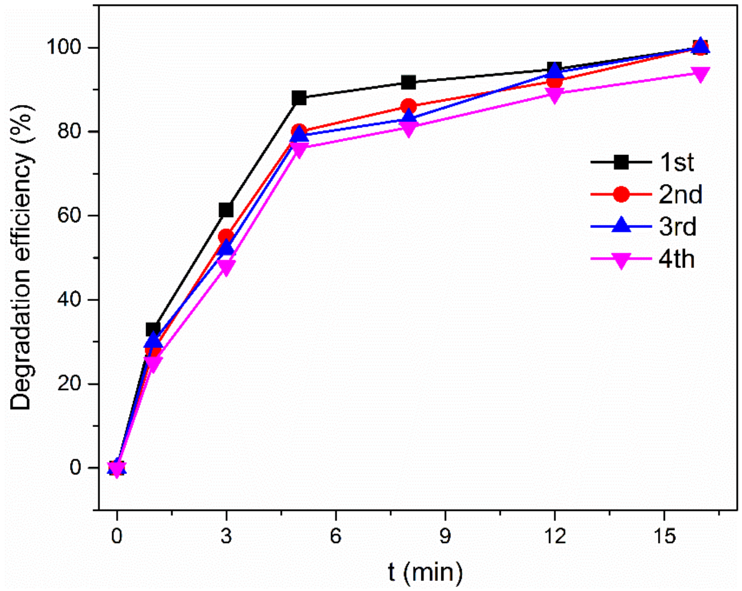



Figure S12. Degradation efficiency of SDZ with $CuFeO_2$ RCs at different catalytic cycles. Conditions: [SDZ] = 8.0 $\mu$M, [$CuFeO_2$ RCs] = 0.1 g $L^{-1}$, [PMS] = 33.0 $\mu$M, and 10 mM phosphate buffer at pH 6.8.

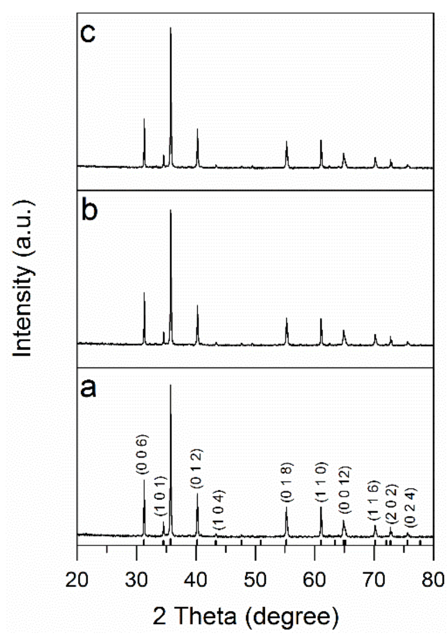



Figure S13. X-ray powder diffraction patterns of fresh $CuFeO_2$ RCs (a) and $CuFeO_2$ RCs after 2nd (b) and 4th (c) catalytic cycles.

## REFERENCES

(1) Guan, Y. H.; Ma, J.; Li, X. C.; Fang, J. Y.; Chen, L. W., Influence of pH on the formation of sulfate and hydroxyl radicals in the UV/Peroxymonosulfate system. *Environ. Sci. Technol.* **2011,** *45*, (21), 9308-9314.

(2) Clifton, C. L.; Huie, R. E., Rate constants for hydrogen abstraction reactions of the sulfate radical, $SO_4^-$. Alcohols. *Int. J. Chem. Kinet.* **1989,** *21*, (8), 677-687.